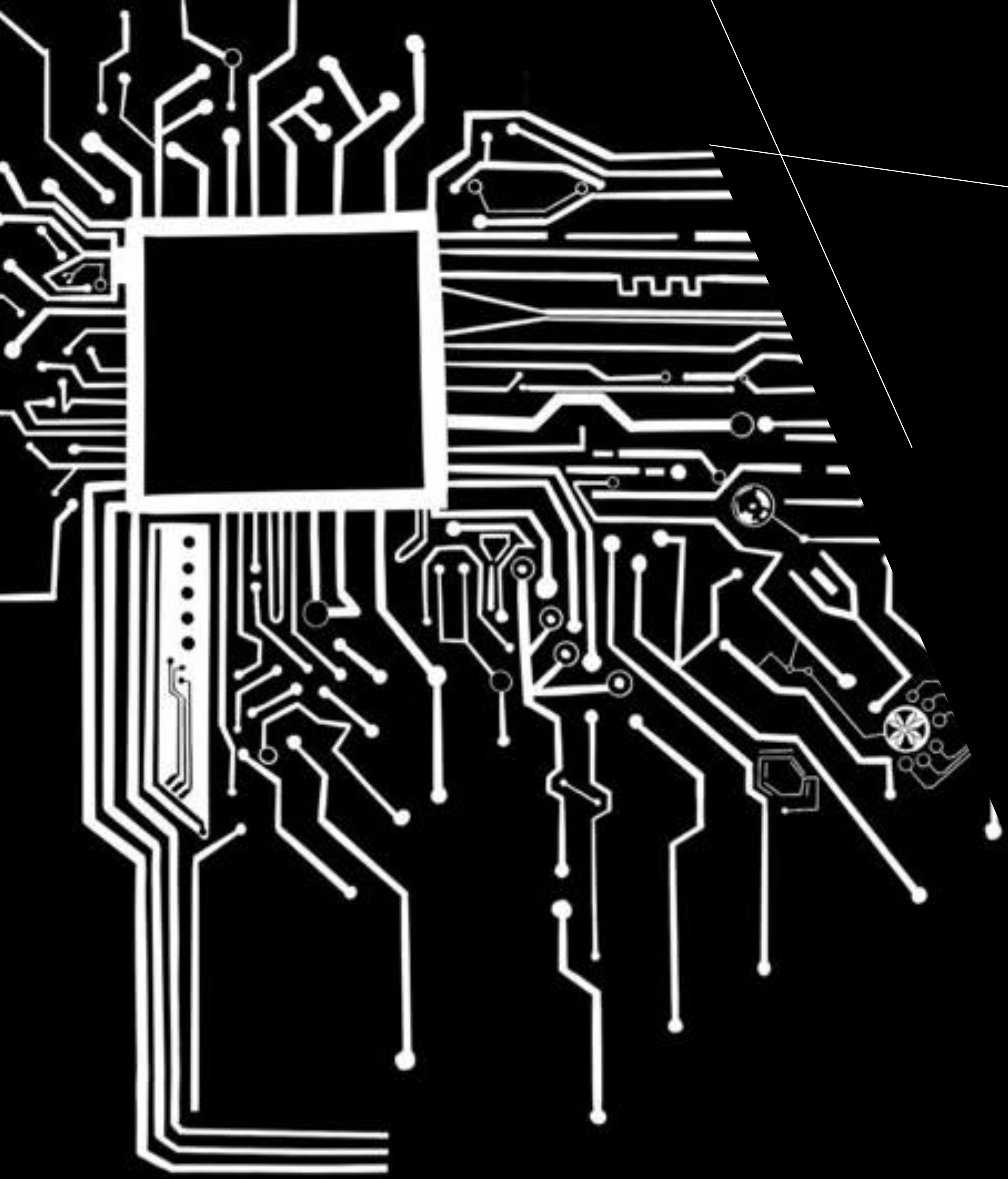

ICE 2343
DIGITAL LOGIC DESIGN

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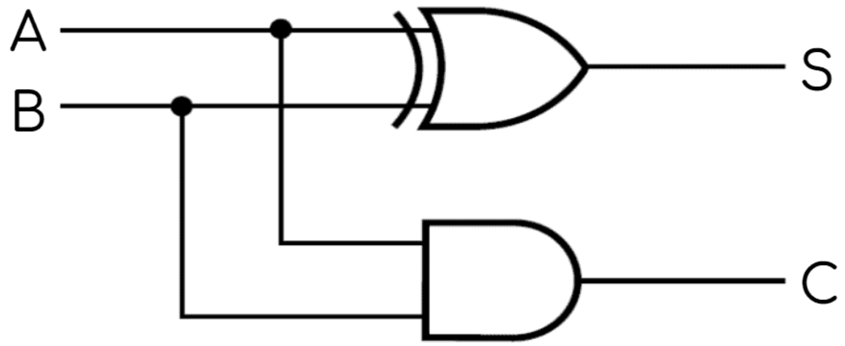
Half Adder

An adder is a combinational circuit that adds inputs to generate output.
Half Adder circuit consists of one AND gate and one Ex-OR gate. The circuit basically has two inputs and generates output in form of a carry and a sum bit.



Circuit Diagram

Sum (S) = $A \oplus B$

Carry (C) = $A \cdot B$

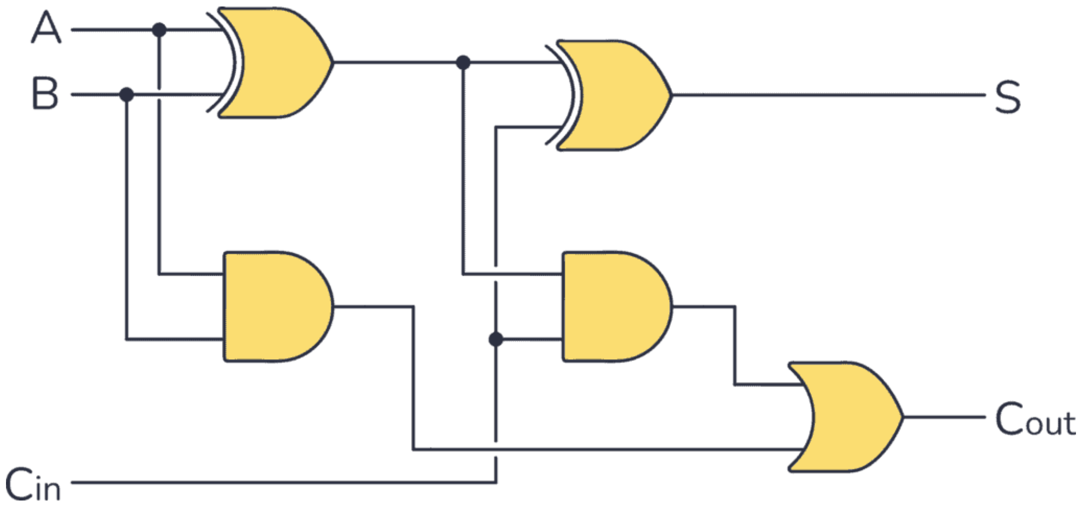
Logical Expression

Half Adder

Input		Output	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adder

A **Full Adder** is designed using two AND gates, two Ex-OR gates and one OR gate.
Full adder is also a combinational circuit meaning it doesn't have memory or storage so, instead it uses additional logic gates to add previous carries to generate a complete output.



Circuit Diagram

$$C_{out} = AB + BC_{in} + AC_{in}$$

$$S = (A \oplus B) \oplus Cin$$

Logical Expression

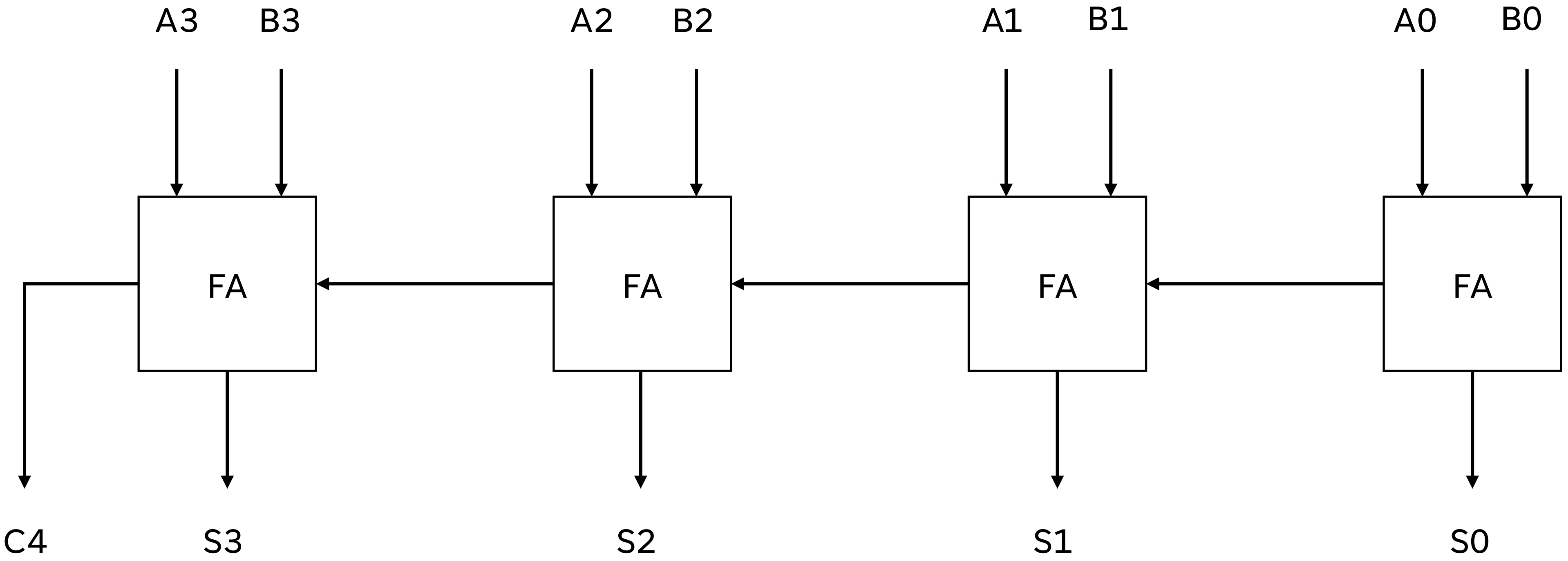
Full Adder

Input			Output	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Problem

1. **Solve yourself:** Implement a full adder using half adder circuit.
2. Discuss the differences between a half and a full adder.

4-bit Parallel Adder



Practice

Sketch 8-bit and 12-bit parallel adder circuit.

Applications

Half Adder:
- Inside full adder circuits
- Addition operation for calculator
- Laboratory trainer kits for learning
- Error detection circuits
- Parity checker

Full Adder:
- Arithmetic logic unit of CPU
- Processor and microcontrollers computation operation
- Counters and timers
- Digital calculators
- Digital signal processing

Reference

Digital Systems Principles and Applications by Ronald J.Tocci

THANK YOU